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Blind by construction: verifying a georeferenced map series when the check shares the error

Three historical map series of Vietnam

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Abstract

Historical map series often supply their own georeferencing control, but agreement with that control does not independently establish placement. We examine this problem through a documented Vietnam Map Archive failure and a controlled reproduction on US Army Map Service L7014 GeoPDFs. Of 510 source files, 62 lacked usable control, 11 failed the graticule check, and 437 entered the faulty build. All 269 sheets with reproduced CRS displacements of 395–528 m (median 455 m) passed that check. The check evaluates registration in the sheet's own geographic CRS and cannot validate the subsequent datum transformation to WGS 84. Comparing the same 715 PDF/PDF edges before and after correction, the number with median outline separation above 100 m falls from 189 to zero. This establishes improved inter-sheet agreement, not independent absolute accuracy: the CRS-selection check and comparison lattice share the adopted Helmert parameters. A second case, from the Service géographique de l'Indochine 1:25,000 series, shows why coherent corner readings can still assign two sheets to the same cell. A third, from the same survey's 1:100,000 series placed automatically from catalogue extents, shows the structure recurring in a pipeline built after the first two were understood: a sheet whose catalogue west and south edges were wrong by 509 m and 544 m passed every per-sheet gate at 0.69% shape error, and was found only because its printed graticule ticks, an input those gates do not use, were read for calibration. Together these cases support a taxonomy of verification scope, committed inputs, detectable faults, and blind spots; the taxonomy is drawn from three cases in one project and is offered as an organizing account of this failure mode, not a claim that it generalizes across archives or pipelines. The contribution is an account, structured for reproducibility, of how checks can remain silent under specific shared assumptions, with explicit separation of source control, build measurements, and serving-state evidence; the derived data and code supporting that reproduction are pending deposit (§11).

Keywords: historical maps · georeferencing · map series · verification · reference frames · reproducibility · spatial data quality

Every number in this file must exist in `figures.md` with a source. Numbers in `figures.md` §5 are quotable nowhere. –>

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1 Introduction

Historical maps are increasingly available as scans, but a scan becomes spatially useful only when it can be placed, queried and compared with other maps. That operation is often described as a georeferencing problem: identify control points, fit a transformation and report an error. For a sheet series, however, the control is frequently already present. It may arrive in a GeoPDF, be printed as graticule intersections, or be recoverable from a repeated neatline and a known index. The difficult work shifts from placing an individual sheet to establishing whether an entire collection has been placed coherently and whether the resulting evidence can detect its own errors.

A datum specifies how map coordinates relate to positions on Earth. Using the wrong datum can move a sheet while preserving its shape. A residual is the difference between two quantities compared by a check; it measures absolute placement only if the comparison can reveal an error in that placement. Figure 1 shows the key mechanism: add the same shift to both quantities and their difference does not change.

This distinction matters because a small residual is not a general certificate of correctness. A check can be accurate for the relation it measures and nevertheless be unable to reveal a datum, frame or adjacency assumption that it shares with the transformation. In the Vietnam Map Archive, we published a substantial subset of GeoPDF sheets hundreds of metres from their intended locations. The pipeline completed, the maps rendered and the per-sheet graticule check passed every displaced sheet. The fault was not hidden by noise; it was invisible by construction because the check compared two quantities translated by the same wrong datum. A current reproduction finds CRS displacements of 395–528 m (median 455 m), with the exact fault population depending on whether it is counted from the fit or from the CRS displacement.

Read the evidence in three steps. Figure 6 locates the affected sheets and shows why some shared edges expose the displacement. Figure 5 counts that disagreement before and after the correction. Figure 7 then asks the separate question of whether sheets reach their expected cells. Together these distinguish a check passing, neighbours agreeing and a sheet being correctly placed.

The size of that silent displacement is a useful warning, but not a benchmark claim. It exceeds the 101 m median error reported by a content-based automatic method and is comparable with some automatic toponym-based results, yet the tasks are not comparable: those methods locate a sheet from content, while our pipeline begins with human-placed or source-supplied control. This paper is about the meaning of verification after control has been obtained, not about claiming that a human-assisted workflow outperforms automatic georeferencing.

We examine three Vietnamese map series with complementary spatial evidence: US Army Map Service 1:50,000 GeoPDFs, Service géographique de l'Indochine 1:25,000 sheets whose corners are printed in grades from the Paris meridian, and the same survey's 1:100,000 second edition, placed from per-sheet catalogue extents and checked against ticks printed along its frame. The archive reports 514 sheets across the first two routes, but that total includes an unverified publication constant (§3). We therefore report each audit's measured population separately. The series setting makes relations available that a single sheet cannot supply: regular lattice

spacing, one-to-one occupancy of cells, shared edges, edition history and a distinction between a proposed correction and what readers currently receive.

The paper makes three contributions. First, it models a survey as cells, archive sheets and institutional printings rather than collapsing those states into one map record (§4). Second, it documents three reproducible georeferencing routes that require no interactive editor: embedded GCPs and neatlines for L7014, detected frames plus printed corner labels for the Indochine 1:25,000, and detected frames plus calibrated catalogue extents for the 1:100,000 (§§5–6). Third, it develops a verification taxonomy (§7). The taxonomy distinguishes checks by scope and by which information they have already committed to. It shows why a lattice residual, a lattice collision, a free seam residual, an outside CRS decision and a tile-key audit detect different classes of fault. It then applies the taxonomy to an automated pipeline we built for the third series, whose own gates share their input with the placement they check (§7.8).

Our claim is deliberately narrow. Series self-consistency is not new, and neither are lattice or adjacency checks. The contribution is to show, with a published failure, that a verification instrument can be predictably blind to one error class by construction. Naming its scope and its committed assumptions is therefore a precondition for interpreting the residual it reports.

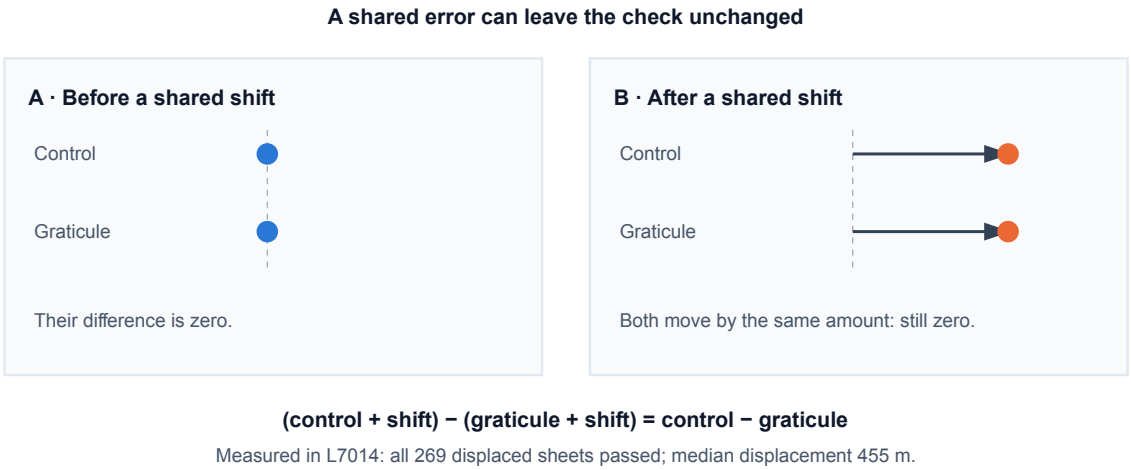


Figure 1: Why a wrong position can pass an internal check. The two panels are a schematic of a shared translation, not measured sheet positions. If both operands move together, their difference stays unchanged. In the controlled L7014 reproduction, all 269 displaced sheets passed the graticule check despite a median CRS displacement of 455 m.

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2 Related work

2.1 Georeferencing a sheet series from its own content

The nearest prior work is Luft and Schiewe (2021), who georeference sheets of the *Karte des Deutschen Reiches* 1:100,000 from their content: they segment the blue water symbols, match the resulting binary mask against OpenStreetMap water rasterised into each candidate sheet's bounding box, take the sheet with the most RANSAC-consistent patch matches, and finish with an ECC registration of content against reference. They report 96% correct sheet identification over 56 sheets and a median georeferencing error of 101 m. What concerns us is not the matcher

but the yardstick. Because a line-preserving transform spreads its error linearly, the extreme always falls at a corner, so they hold the transformed corners against "their expected positions in the series' sheet layout", and they state the consequence in print: "the alignment of corners also directly determines the ability to seamlessly join neighbouring transformed map sheets, which is a major concern for map users." Using the series' own lattice as the standard against which a sheet is judged is therefore established practice. The check we describe in §7.2 confirms it on a second series and a different frame; it does not introduce it.

Two properties of that construction bound what the yardstick can see, and both are this paper's subject. The first is that the sheet-layout bounding boxes are prior knowledge — "constructed in advance from the map series' metadata" — and are used twice: once to give the output image its spatial reference, and once as the ground truth the transformed corners are measured against. The number that results is the residual of the image registration *inside an assumed frame*. Should the frame itself be wrong — the datum, not the alignment — both sides of the comparison move together and the residual does not move at all. Luft and Schiewe are explicit that the corner metric is designed to exclude one class of error: corners are used because "neatlines are the first thing constructed and have the least projection error, [so] they can be assumed to be drawn at the 'correct' place", which keeps surveying and drawing error out of the measurement. It keeps datum error out of it too. §7.4 reports a several-hundred-metre datum fault across part of the 437-sheet L7014 denominator, which a check of exactly this shape passes without a single rejection.

Luft and Schiewe say as much about a third study, and the remark is the clearest statement of this paper's thesis we have found in the prior literature. Comparing their own result with Burt, White, Allord, Then and Zhu (2019), who georeference USGS topographic sheets from neatline corners and graticule intersections, they note that those authors reach "almost perfect accuracy of 1–4 px RMSE" and immediately qualify it: the figure "reflects the detection accuracy of graticule intersections on the map image and is hardly influenced by the georeferencing process itself." That is precisely the distinction Table 1 formalises — a residual reports the relation it measures, not the one its reader wants — observed in passing rather than developed. We take it as a precedent for the argument, not as a gap.

The second is that the seam is asserted rather than measured. Corner displacement is computed per sheet against the layout and averaged over four corners into a single value; no two neighbours are ever compared to each other. A per-sheet residual cannot separate a sheet that sits slightly off its cell from two sheets that claim the same cell — the failure that caught `Ha Noi`, 75 km from the city it is named for while satisfying every per-sheet check (§7.2). Inter-sheet agreement as a *constraint* is well established: Janata and Cajthaml (2020) adjust a 250-sheet series under explicit adjacency conditions. We treat unforced inter-sheet agreement as a diagnostic in this archive, without claiming priority for that use.

Gede and Varga (2021) are the closest prior route to the Indochine method in §6. They detect the map-content corners, recognise a sheet identifier by OCR, derive the quadrangle from that identifier against a known layout, and use the corners as GCPs. Their geometric analysis automatically filters false corner detections. Our sheets differ at the decisive step: they print their own corners in grades from Paris and have no identifier-to-extent table, so the coordinate reading replaces the lookup. Uhl et al. (2018), in turn, use GCP displacement vectors as a per-sheet anomaly measure across map archives. Both methods reinforce the point of this paper: a useful per-sheet diagnostic is not thereby a diagnostic of a fault that belongs to the series frame.

Heitzler, Gkonos, Tsorlini and Hurni (2018) belong to the same graticule-intersection family as Burt: they locate grid intersections with a Hough transform and warp each coordinate cell

independently rather than the sheet as a whole, reporting per-module precision on 20 sheets — 100% corner detection, 94.6% grid-intersection placement, 95.6% coordinate interpretation — but no ground-distance error anywhere in the paper. Their only demonstration that the result improves on the existing georeferencing is a single adjacent-sheet pair: a stream's topological break across the seam closes under their method while a road's mismatch persists in both, and they note that the latter needs conflation rather than better georeferencing. One pair, chosen, unmeasured — but it is the same reflex §7.3 turns into a census: the seam is already the evidence a reader reaches for, before anyone counts it.

2.2 Inter-sheet agreement, spent as a constraint

Inter-sheet agreement has been used before, as a constraint. Janata and Cajthaml (2020) adjust 250 sheets of the First Military Survey of Bohemia in a single least-squares solution — 6,849 control points, more than 27 per sheet — in which the coincidence of shared sheet edges enters as a set of condition equations, robustified by iteratively reweighted least squares under Huber's M-estimate. The series was mapped *à la vue* and has no geodetic basis, so no projection can be inverted and the sheet edges are the only geometry available a priori; using them is the right move, and their best solution reaches 280 m RMSE on a corpus whose own drafting is far coarser than that. The consequence for verification is structural rather than accidental. Once edge identity is imposed as a condition, "the adjacent edges fit exactly together" by construction, and the seam residual is identically zero however the mosaic as a whole is placed. Information spent as a constraint cannot be spent again as a diagnostic. The seam census we report in §7.3 is legible only because no adjacency condition was applied.

This is the same shape as the limit described in §2.1, arrived at by a different route, and neither is a defect in the work that exhibits it: Luft and Schiewe cannot see a frame error because the frame stands on both sides of their metric, and Janata and Cajthaml cannot see a seam error because the seam is an input to their solution. What the two have in common is that the quantity a reader would most naturally reach for as evidence of correctness is, in each case, the quantity the method has already committed to. We take this as the general statement of the problem rather than as a gap in the literature.

Both papers also report the limits of their own instruments, which is the register this one adopts. Janata and Cajthaml assembled about fifty control points, held out from the adjustment, to decide whether their mosaic or the Mapire.eu layer was better placed, and reported that "it is not possible to unambiguously decide which dataset is better adjusted" — the medians agree at about 250 m and fifty points do not cover Bohemia. They separately report a null on weighting control points by object category. We report comparable nulls in §8.

2.3 Per-sheet consistency checks

Meijers and Schoonman (2025) reach the neatline the way we do and applies a substantial per-sheet check set. MapEdge detects the frame of 557 sheets across three 1:50,000 series by taking 1D black-pixel profiles along each axis of a low-resolution overview, ranking candidate peak pairs by a fuzzy metric over separation and expected position, then repeating the profile on full-resolution edge strips cut into patches, fitting a line per side by RANSAC and intersecting the four lines. The corners serve as both the control points and the mask, at about 15 s per sheet. Both that method and ours read the same quantity off the same strips; the difference is peak selection, where MapEdge ranks candidates against a user-declared fuzzy width prior and we take an unthresholded `argmax` because on these sheets the thick neatline is the darkest

thing on the strip by a factor of three. Ours needs no prior and cannot be tuned; theirs adapts to layouts ours would fail on.

What matters here is the battery of five checks applied afterwards: sufficient rim samples and RANSAC inliers; an internal test that opposite sides and diagonals agree; an external test of documented sheet size in centimetres times scan dpi against the measured pixels; the distribution of inlier perpendicular distances, which detects paper bulging or caving; and a visual contact sheet of every detected corner. The battery works — on three sheets it revealed that sheet-size exceptions had been digitised wrongly in the authors' own sheet index, which was corrected as a result. These checks concern per-sheet geometry. Our §7.2 example has internally coherent readings while occupying another sheet's cell. We did not run the MapEdge implementation on that sheet; the comparison concerns the scope of internal geometric checks, not an observed MapEdge verdict.

Meijers and Schoonman also state in print the failure our own detector hits: "linear features, such as graticule lines, can occupy the same number of black pixels as the neat line." §6 reports a worked case of exactly that — a walk stopping on the graticule band, 40 px and 170 m short, every edge rejected — and what we do about it. We take this up as the open problem its author names, not as a deficiency we found.

2.4 Verification beyond geometry

External evidence is also used in automatic georeferencing, but it answers a different question from the internal checks above. Milleville, Verstockt and Van de Weghe (2022) geocode recognised toponyms, filter candidate locations and score predictions against independent official metadata polygons or GeoTIFF corners. Their reported mean errors are 316 m and 287 m on two adjacent-sheet corpora. Their own caution is directly relevant here: centre error can agree while the predicted extent remains wrong. An apparently simple metric is only evidence for the property it actually measures.

Text and image understanding methods extend the available evidence but do not remove that constraint. Wijegunarathna, Stock and Jones (2025) report approximately 1.03 km mean error for locality georeferencing from textual descriptions; Namgung and Chiang (2022) use spatial relationships among words to improve post-OCR rather than treating a map as a line of prose. mapKurator (Kim et al., 2023) and the ICDAR 2024 MapText competition (Li et al., 2024) demonstrate the scale at which map text can now be detected, recognised and linked. These are important routes toward independent reference, but they are not substitutes for stating what a geometric check can and cannot observe. We therefore use outside information only where it separates competing datum hypotheses (§7.5), rather than presenting it as a universal ground truth.

Recent multi-sheet work makes the same distinction from another direction. Kuna, Panecki and Zawadzki (2024) identify a difference between the mathematical framework and topographic content as a source of error accumulation during mosaicking; they caution that, despite good geometric parameters, a grid without the correct datum relationship is "not a reliable element of the mathematical framework". They then rectify all 60 sheets of their corpus against an independently generated grid. Xu, Li and Lv (2026) likewise construct an external reference grid to supply conjugate control points for batch georeferencing. These studies do not make our blindness claim, but they show why the independence of a frame cannot be assumed from a low internal residual: it is a design choice that has to be stated. In a related operational setting, Ingensand, Lecorney and Blanc (2022) describe validation indicators based on GCP count, computed error and GCP coverage. Those indicators are useful gates; Table 1 specifies

the error classes they leave outside their scope.

Tyagi and Dubey (2026) read printed map text with OCR and a multimodal model to derive a sheet's extent, which is the closest published neighbour to the printed-corner method in §6. We have not obtained the full text and therefore attribute no result to it. Legend detection is likewise outside this paper's scope: it concerns what a sheet depicts, not whether the sheet's spatial frame is independently verified.

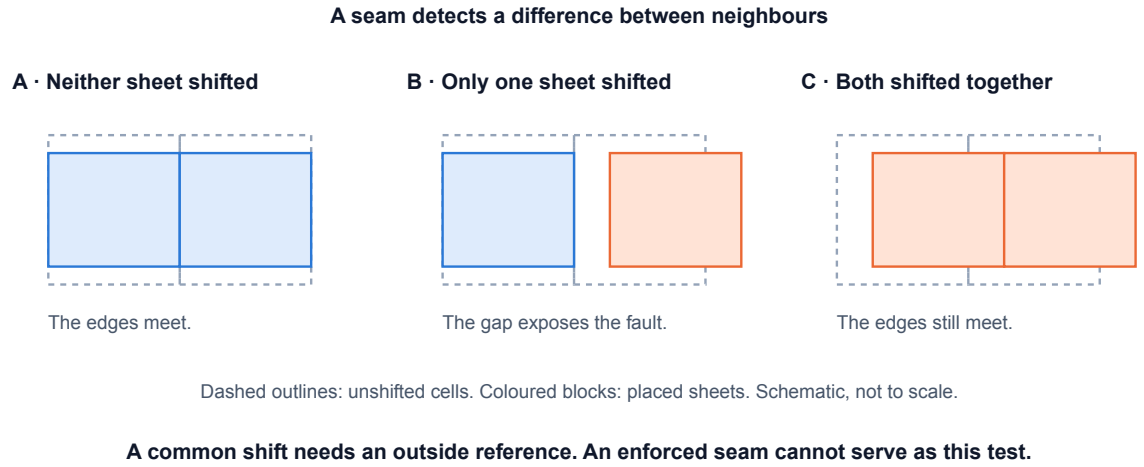


Figure 2: When a seam can reveal a displacement. Schematic, not to scale. A gap opens when neighbouring sheets move differently (B); a common translation preserves their join (C). The census is diagnostic only when adjacency was left free during the warp. Figure 6 shows the geographic populations that make case B possible in this archive.

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3 Corpus

The corpus comprises one Cold-War and two colonial Vietnamese sheet series. L7014 is the US Army Map Service 1:50,000 coverage of Vietnam, held as GeoPDFs by the Perry-Castañeda Library at the University of Texas at Austin. The second is the Service géographique de l'Indochine 1:25,000 coverage of Tonkin and Thanh Hóa, held by Cartomundi at Aix-Marseille Université/CNRS. The archive previously reported a combined total of 514 sheets, comprising a 452-sheet L7014 publication count and 62 pipeline-georeferenced Indochine sheets. Because the 452 is a hand-maintained publication constant that cannot be checked against the missing serving manifest (§7.6), the sum is an archive-reported total, not a verified count of processed or currently served sheets. The measured populations below supply the denominators for the results. The L7014 material dates from the 1960s–70s; the Indochine material from 1903–1927. The Indochine sheets are colonial-era survey products of the Service géographique de l'Indochine, and their control and coverage reflect the administrative priorities of that survey rather than a neutral sampling of the territory; we make no claim about the map content beyond the geometric properties used for verification here.

The third series is outside that total. The Service géographique de l'Indochine 1:100,000 second edition (1947–1959) is held by Cartomundi as **492** catalogue records, each describing the eastern or western half of a numbered cell, with a per-record extent. The archive holds **360** of these halves as scans, each matched to exactly one record. **446** of the 492 records declare a longitude

and latitude span inside the series' standard band (0.36–0.43 and 0.48–0.53 grades); the other 46 are catalogue anomalies, handled separately. At the recorded state, **314** of the 360 are public and georeferenced (§7.8).

Several L7014 denominators appear below and they are not interchangeable; Figure 6 shows all of them on one map; Figure 3 separates the audit populations. The series index names **627** cells. **510** of those are held as GeoPDFs. Of the 510, **448** reach the graticule check after 62 files are excluded for missing control. The check rejects 11, leaving **437** successfully warped files in the faulty build, and it is this 437 that the datum measurements in §7 are taken over. A further **24** sheets are georeferenced by hand rather than from embedded control, and sit outside that fault. The **452** in the published mosaic is a hand-maintained constant rather than a count read back from the serving manifest, and a corrected rebuild produces 436; we report it as the archive's own claim about itself and flag it in §7.6 rather than silently substituting a number the live layer cannot confirm. On the Indochine side, **79** cells are indexed and **75** held; **62** were georeferenced by the pipeline, **58** were read independently for the lattice check in §7.2, and the IIIF survey in §7.7 covers the **83** image sources those sheets resolve to, some cells holding more than one edition.

The first two series are deliberately unlike in how they supply spatial evidence. L7014's source PDFs carry GCPs and a CRS declaration but require a test of that declaration. The Indochine sheets carry no embedded georeference but print corner coordinates and a regular quadrangle scheme. The 1:100,000 series sits between them: it prints graticule ticks rather than corner labels, and its catalogue supplies an extent for every sheet, so its placement comes from outside the sheet and its print is read only to calibrate that source. This contrast is useful for the paper's question: the checks in §7 do not depend on a single ingestion route, but on relations a sheet series makes available after its coordinates have been read.

The corpus is not a representative sample of all historical maps of Vietnam, nor is it a ground truth dataset for absolute positional accuracy. Its holdings, digitisation quality and availability follow the decisions of the contributing institutions. The findings below should therefore be read as a failure taxonomy demonstrated on two well-structured series, not as a benchmark ranking against methods that recover a sheet's location from no prior spatial information.

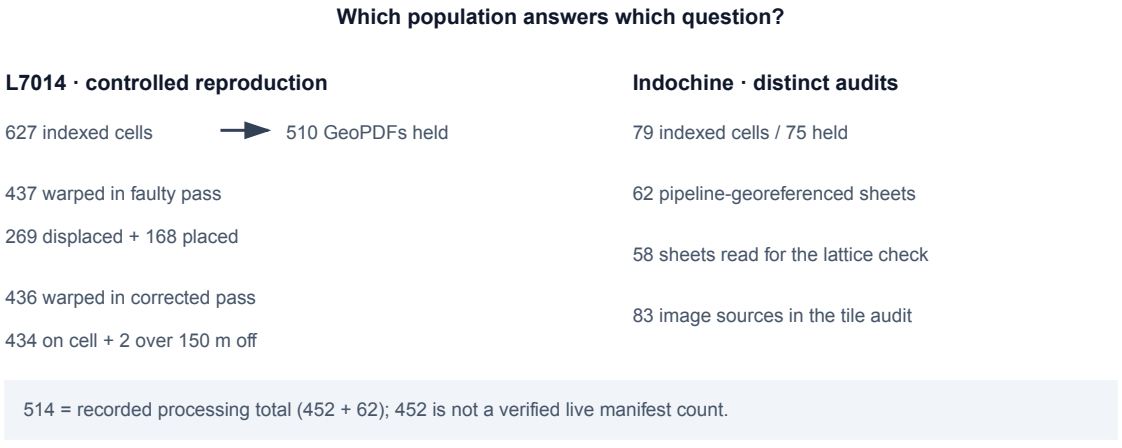


Figure 3: Choose the denominator that belongs to the question. The L7014 reproduction, Indochine audits and historical processing total count different populations. "Placed" uses the CRS-displacement classification; "on cell" uses the fit check. They must not be treated as the same test. The 452-sheet publication constant is not verified against the live manifest.

4 C1 · Modelling the survey

The archive models three related but non-interchangeable things: a survey cell, a sheet the archive can serve, and a physical printing held by an institution. A cell is the quadrangle named by a series key and sheet number. A served sheet records the archive's present route to that cell, whether as a `maps` row, a raster in a pre-tiled mosaic, or an unheld gap. A printing records an edition at a particular institution, including its title, date, part and source identifier. The distinction is not abstract: the same cell can be reprinted years apart, renamed, or issued as eastern and western halves before a later assembly. The three-way split echoes the FRBR work/expression/manifestation/item separation long used in archival and library cataloguing (IFLA Study Group on the Functional Requirements for Bibliographic Records, 2009); we adopt a coarser, three-entity version of that precedent because the archive's verification relations operate at the cell and served-sheet levels, not at FRBR's finer expression grain.

This model makes the denominator visible. The ordinary `maps` table contains only successful archive objects, so it cannot distinguish a survey of nine sheets from a survey of 627 cells of which nine have been acquired. A series index supplies that denominator. It also prevents duplicate editions from being counted as extra geographic cells, while retaining their bibliographic differences at the printing level.

Status is derived rather than stored: whether the archive holds or serves a cell follows from the current route and source records. Likewise, the printing index has no foreign key to the archive's own sheet table. An institutional printing for an as-yet unindexed cell is precisely the discovery the index exists to record; rejecting it because the archive does not already possess the cell would turn new catalogue knowledge into an import error. The same separation lets the dry run in §7.6 validate a corrected representation before the serving mosaic is changed, and it supplies the adjacency and occupancy relations required by the seam and lattice checks.

5 C2 method A — L7014

L7014 is a case where georeferencing control already arrives with the object. Each US Army Map Service 1:50,000 GeoPDF carries ground control points and a printed neatline. We convert the control points into a VRT, clip the image to its neatline, warp the clipped image to Web Mercator, and build the resulting rasters into one PMTiles archive. A compact companion index holds an outline, sheet number, name, edition and date for each constituent sheet. The choice of a pre-tiled archive is a serving decision: it turns a large series into one static object-store resource rather than a tile server or hundreds of per-view requests.

The procedure is intentionally conservative about what the source file establishes. Embedded GCPs locate the raster under the source's CRS claim; the neatline determines what part of the scan is map rather than collar, legend or paper. Neither fact verifies the CRS claim, and neither independently tests a join after the warp. The rest of this paper follows from taking those limits seriously. In particular, the datum fault in §7 was not a failure to find control points or to execute a warp. It was a failure to test an assumption carried by both.

Figure 4 shows what a single sheet makes available, and why those two facts are not independent evidence about each other. The sheet prints its own corner coordinates and, in its collar, the datum those coordinates are expressed in. A check that reads control points under the declared datum and compares them with the printed graticule is therefore comparing two quantities that a datum error moves together — the mechanism §7.4 measures.

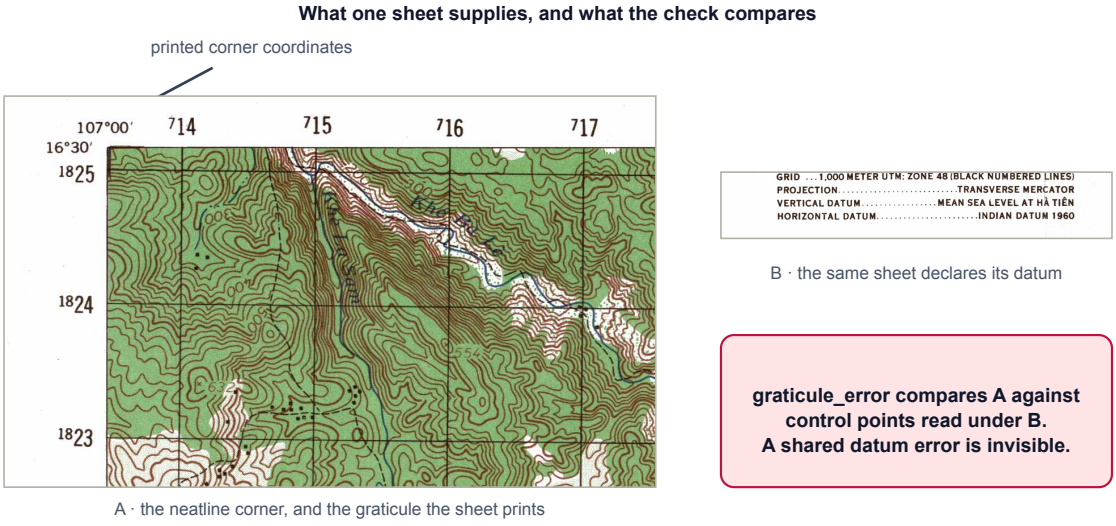


Figure 4: The source of the shared assumption. The graticule and the control points are interpreted in the sheet's declared datum. Their agreement can test internal consistency but cannot independently establish that datum. This correctly placed sheet illustrates the inputs, not an example of the measured displacement. A Lưới 6441-4, AMS Series L7014 (US Army Map Service; scan by the Perry-Castañeda Library, University of Texas at Austin). US Government work, public domain.

6 C2 method B — Indochine

The Indochine 1:25,000 series has no embedded georeference, but it prints the required information. On each sheet, four corner labels give longitude and latitude in grades from the Paris meridian. We detect the pixel positions of the inner neatline, OCR the four short labels just inside its corners, convert grades to degrees by multiplying by 0.9, add 2.3372 degrees for the Paris-to-Greenwich offset, and write the resulting four control points as a Georeference Annotation. On Như Trác, for example, the printed box converts to 18.75 by 12.48 km with aspect ratio 1.50, while the detected neatline measures 4,496 by 3,014 px with aspect ratio 1.49. These are independent checks on the reading, not a claim that four corners exhaust the sheet's geometric uncertainty.

The detector differs from a simple inward edge walk because the printed layout has several plausible linear features. From the paper edge inward sit a thin line, a thick neatline, another thin line, blank paper, and a graticule band. An inward walk stops at the graticule band: on Như Trác, 40 px or about 170 m short, with 14–41 px residuals and all four edges rejected. We instead fit the thick neatline. It is the darkest feature in the edge strips by a factor of three, so its per-patch choice is an unthresholded `argmax`; the fitted line then supplies both the side position and the scan's rotation. The rim is only sought after de-tilting and averaging a narrow strip along that line.

Two implementation constraints follow from the scans rather than from an aesthetic preference. The rough pass averages a narrow band, not half a sheet: scans can be up to 0.8 degrees off square, and a half-sheet average smears a ten-pixel line across 30 pixels, moving Như Trác's top neatline by 35 px. And the rotation is solved once from all four sides. Independently fitted side slopes differ by only about a thousandth, but over several thousand pixels that is

enough for opposite sides of a quadrilateral to differ by 14 px where the projection predicts four. One global angle preserves a single sheet geometry; each side subsequently retains only its own offset.

The method does not conceal its remaining ambiguity. The rim is a pair of thin lines seven pixels apart. We select the inner line because it gives 0.29% disagreement between the two ground-scale estimates, rather than 0.49% for the outer line, and because it bounds the drawn map. That preference is below the roughly 0.3% floor set by paper shrinkage and detection noise. Reading the printed graticule ticks would resolve it and would supply interior control points; it is deferred rather than represented as an achieved precision.

The 1:100,000 series. The second edition does not print coordinates at its corners. It prints graticule ticks along all four sides, labelled every 0.1 grade (0.2 on northern sheets), and its catalogue supplies an extent for every half-sheet. We therefore place each sheet from its catalogue record rather than from its own print. The pixel frame is detected as in the 1:25,000 case, with constants re-measured for this series: an outer decorative frame sits about 175 px, roughly 1.3 km on the ground, outside the neatline that bounds the map. The four detected corners are then tied to the record's extent, corrected by a systematic catalogue-to-print offset.

That offset is the one place the print enters. On a 16-sheet calibration sample, stratified by latitude, we located the printed ticks against the detected neatline and read their values. Seven sheets gave complete observations. A fit linear in latitude left 235.2 m mean and 741.5 m maximum residual, failing a 250 m gate; the maximum was a single sheet, Tri Binh (W). With it excluded, six sheets spanning 10.5–20.9°N fit at 76 m mean and 188 m maximum, and that fit was accepted. The exclusion was recorded at the time as a judgement made to unblock the series, not as a finding. §7.8 reports what it turned out to be.

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7 C3 · Verification

7.1 What a series already knows about itself

The lattice, the seams, the rim constant and the sheet numbering are all prior knowledge a series carries about itself, and each supports a check. The checks do not overlap the way their surface similarity suggests. **Table 1** states this paper's thesis: rows are checks, columns are error classes.

Table 1 — which check sees which error class. Legend: ✓ can flag it under the stated inputs · · not established as a diagnostic here · — not applicable · ∅ the specified fault cannot disturb the constrained or shared-input comparison · † only where the fault is *partial* across the series, so that a displaced sheet borders an undisplaced one; a displacement common to every sheet leaves every seam closed (§7.3).

check	scope	misregistration inside the frame	the frame/datum itself wrong	two sheets on one cell	sheet-index entry wrong	serving/tile fault
Corner residual vs sheet layout — *Luft & Schiewe 2021*	per-sheet	✓	·	·	·	—
Internal consistency: sides, diagonals, cm×dpi, paper bulge — *Meijers & Schoonman 2025*	per-sheet	✓	·	·	✓	—
`graticule_error` — sheet's GCPs vs the graticule it prints	per-sheet	✓	∅ ‖	·	·	—
Adjacency as a least-squares constraint — *Janata & Cajthaml 2020*	inter-sheet	✓	·	·	·	—
…the same adjacency read back as a residual	inter-sheet	∅	∅	∅	∅	—
Seam census (adjacency left free)	inter-sheet	✓	✓ †	✓	·	—
Lattice residual	series	✓	·	·	✓	—
Lattice collision	series	·	·	✓	✓	—
`pick_crs` — registration points scored against the adopted cell (§7.5)	per-sheet + index	·	✓ ‡	·	·	—
Span and frame gates vs catalogue box (§7.8)	per-sheet	✓	·	·	✓ §	—
Line-fit residual (§7.8)	per-sheet	∅ ¶	·	·	·	—
Printed ticks vs catalogue (§7.8)	per-sheet	·	·	·	✓	—
Tile-key audit	serving	—	—	—	—	✓

‡ Detects disagreement with the adopted frame when a cell score is available. The candidate and reference cell share Helmert parameters; the decision does not validate those parameters against independent absolute control. § Only when the index error changes the box's shape; an error that enlarges both axes by similar proportions passes (Tri Binh, §7.8). ¶ For the specific case of a boundary pinned to the wrong printed line: a ruled interior line fits straighter than a hand-engraved neatline, so the residual cannot flag the choice. ‖ For a datum *translation* only; an ellipsoid error is detected (§7.9). The L7014 cells of the four checks this paper runs (`graticule_error`, lattice residual, `pick_crs`, seam census) are tested by fault injection in §7.9. Every other entry describes diagnostic scope, not measured detection rates from running the published methods on this corpus.

Three cells carry the argument, and each is sourced to a different paper's own design.

First, `graticule_error` on "frame/datum wrong" is ∅, not ·, for a datum translation. It does not merely miss the fault; it cannot report it under those shared-input conditions. The sheet's control points are read into the sheet's own datum and compared against the graticule that same sheet prints, so a datum error moves both sides together — the check passed all 269 displaced sheets, and returns 2.167e-12 on a sheet it places correctly (§7.4). Luft and Schiewe's corner metric has the same shape, and says so without meaning to when it justifies corners on the grounds that neatlines "can be assumed to be drawn at the 'correct' place".

Second, adjacency-as-constraint turns an entire row to ∅. Once edge identity is a condition equation, the edges fit by construction; the information was spent. Our seam row is the same measurement left unspent, and it is the row that caught the fault (§7.3).

Third, the internal per-sheet checks listed here do not establish unique occupancy. The `Ha Noi` example has coherent internal readings but conflicts with another sheet's cell assignment (§7.2). This is a scope argument, not a measured result from running MapEdge on `Ha Noi`. A check supplied with an independently verified sheet-to-cell assignment could detect a misplaced sheet individually; internal geometry alone cannot establish that assignment.

The claim this table licenses:

> *A verification instrument can remain insensitive to an error shared by its inputs or removed from its residual by an imposed constraint. Interpreting its residual therefore requires stating the comparison, the committed assumptions, and the faults capable of disturbing it.*

It does not claim that the lattice is novel, that adjacency is novel, or that any of these papers made a mistake. Every ∅ is the ordinary consequence of a sound design — including ours, which is the worst row in the table: `graticule_error` was shipped, trusted, and silent for the affected subset of the 437-sheet denominator.

7.2 The lattice: residual *and* collision

We read 58 sheets independently, before asking whether their readings formed a series. Their west edges contain nine distinct values with a smallest step of 0.200 grades; their north edges contain fifteen with a 0.125-grade step. Both sets lie on those lattices with residual 0.000 at the precision of the printed corner readings. A separate pixel-space check reaches the same conclusion about the scan geometry: the rim-to-neatline offset has a median of 84.2 px, and all 58 sheets lie within 15% of it. Sheet numbers introduce no row-order exceptions. These are

relative checks. They establish that the readings make one internally coherent grid; they do not establish where that grid lies in a modern datum.

The lattice has a second operation that a residual cannot perform: it makes duplicate occupancy visible. `Ha Noi` (sheet 20, 1903) is the useful counterexample. Its corners agree with one another, its ground scale agrees with its pixels to 0.78%, its rim is at the series offset, and its content is unambiguously Hanoi. Yet its printed latitude places it 75 km south of the city, on the cell occupied by `Ninh Bình`; the longitude is right and the latitude is a clean 0.75 grades out. These observations establish the limits of internal consistency, without claiming an observed pass from the MapEdge implementation described in §2.3. Nor would a small lattice residual: a sheet can sit perfectly on a cell that belongs to another sheet. The collision is the evidence. A collision is a review flag: it may represent intentional alternative editions, a printed index anomaly, or an archive/transcription error. Both editions are retained here pending a human decision about which explanation applies.

Residual and collision therefore answer different questions. The former asks whether independently read coordinates form the expected spacing; the latter asks whether the sheet index is one-to-one. Conflating them would make the `Ha Noi` case disappear into an otherwise excellent grid.

7.3 The seam census

The census measures outline agreement rather than continuity of roads or other map content. The lattice identifies shared edges but does not constrain the PDF outlines. For each edge, `l7014_seams.py` samples 25 equally spaced positions between 10% and 90% of its length, avoiding corner junctions. At each position it finds the nearest point on each outline and measures their separation using a local longitude/latitude-to-metre approximation. The edge statistic is the median of these distances; the reported census median is the median across edges. The 100 m and 300 m cutoffs summarize the observed tail, rather than define independently calibrated acceptance thresholds. For nearly straight parallel edges the distance principally reflects normal separation; it need not recover along-edge displacement or a full displacement vector. The 25-point, 10–90%-of-length sampling design was fixed by inspection rather than swept for sensitivity to sample count or window; we report this as an open parameter rather than a calibrated choice, and expect the census's bin counts, not its exact median, to be the figure robust to that choice.

The lattice compares sheets to the series frame. A seam compares each sheet to its neighbour while leaving that relation out of the warp. In the mosaic-only reproduction, 717 PDF/PDF seams have a median displacement of 9.2 m and 97 exceed 300 m. Splicing in the separate hand-sheet rows produces a second, hand-extended cohort of 778 seams: 36 additional JPG/PDF seams and 25 JPG/JPG seams. The hand-extended cohort has the same 9.2 m median and 111 seams over 300 m. The hand-extended cohort includes the mosaic-only cohort; the two denominators are reported separately rather than treated as independent samples.

The distribution is not a diffuse consequence of paper distortion. The 36 hand-to-mosaic seams span 1.8–446.2 m; none falls in the earlier 447–504 m band. A seam reports only the component normal to the shared edge: the hand comparisons have medians of 438.0 m east/west and 133.7 m north/south, recombining to 457.9 m, close to the 454.9 m median CRS displacement.

The hand-sheet comparisons must be distinguished from the free PDF/PDF comparisons. The hand workflow supplies lattice-derived ground corners, and the hand manifest is assembled from those ground coordinates rather than independently measured warped image boundaries. A

JPG/PDF join therefore compares a PDF outline with a lattice-derived boundary. A JPG/JPG join compares two such boundaries. The 25 JPG/JPG distances of 0.0–0.6 m describe consistency of those ground coordinates, not an independent validation of either image registration. Only the PDF/PDF cohort is used as free-seam evidence here.

The two builds differ in the CRS-selection flag and in which files survive the checks. The faulty build contains 717 PDF/PDF seams, of which 189 exceed 100 m and 97 exceed 300 m. The corrected build contains 715, none above 100 m (Figure 5). To separate change in geometry from change in membership, we matched edges by their unordered sheet-ID pair. All 715 corrected edges occur in the faulty census. On these same edges, 189 fall from above 100 m to at most 100 m, with no crossings in the opposite direction; the maximum corrected edge median is 72.4 m. The two faulty-only edges measure 3.6 m and 15.6 m, so their exclusion cannot explain disappearance of the large-displacement tail. Matched-cohort medians are 9.2 m before and 6.8 m after correction.

These are descriptive comparisons within one series. Edges share sheets and are not independent replicates; their distances do not measure absolute accuracy or content alignment. The matched analysis, input hashes and excluded pairs are reproducible with `scripts/l7014/paper-seam-pairs.mjs`.

This is also the limit of the instrument. What the seam census detects is *differential* disagreement across a shared edge, and it fired here only because the fault was partial: a displaced sheet was somewhere adjacent to an undisplaced one. Had every sheet in the series moved by the same vector, every seam would have closed exactly as before and this census would have reported nothing. The instrument found the fault because the fault had a boundary — the boundary Figure 6 maps.

The seam anomaly was the first inexpensive indication, in this archive, that it had split into two spatial populations. The measurement needs neither a reference map nor a new set of control points: it needs only two sheets that claim to share an edge. Its value is diagnostic: it catches a disagreement that a per-sheet metric cannot see, provided adjacency has not already been imposed as a condition of the solution. A common displacement preserves a seam, just as it preserves any other relative measurement.

A single seam has been offered as this kind of evidence before (§2.1): Heitzler et al. (2018) show one adjacent-sheet pair closing under their method, by eye, with no threshold and no denominator. What changes here is the count — 717 free seams, binned and thresholded, not one pair chosen to illustrate.

7.4 The blind self-check

The seam anomaly led to a datum fault in the L7014 GeoPDF pipeline. The 437-sheet denominator contains a fault population that depends on the measurement: 276 sheets under the fit criterion and 269 under the CRS-displacement criterion. Their displacements span 395–528 m, with a median of 455 m. Some files carried an NGA LGIDict datum label that GDAL could not map and were warped after a warning with a WGS 84 fallback. More subtly, even a sheet explicitly labelled `Indian_1960` could fail: the `EPSG:4131` to `EPSG:4326` operation returns the input unchanged outside its area of use. The behaviour is non-uniform: a probe at 106.00 E, 16.00 N moves about 470 m, while one at 109.25 E, 13.25 N moves 0. The production archive therefore contains a discontinuity rather than one uniform translation.

Our existing `graticule_error` check did not report it. In the faulty pass, 448 files reached the check after 62 were excluded for missing control. It rejected 11 and retained 437, including

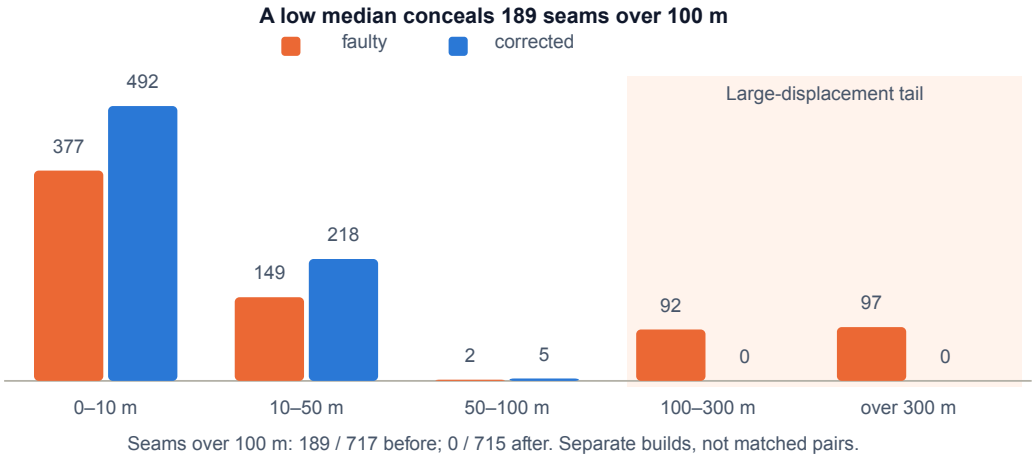


Figure 5: The large-displacement tail is absent after correction. Counts of free PDF/PDF seams in each displacement bin. The faulty build has 189 of 717 seams over 100 m; the corrected build has 0 of 715. These are separate builds, not a paired analysis of the same 717 seams. A matched analysis of the 715 common edges also gives 189 versus zero above 100 m (§7.3). A small median (9.2 m before, 6.8 m after) alone would obscure the faulty build's large tail.

all 269 with reproduced CRS displacements of 395–528 m. This is not a sensitivity limit: the rejections it did issue fire between 0.004 and 0.076 degrees off the printed graticule, so the instrument has a working dynamic range and still returned nothing on a 455 m translation. On A Lưới — which `pick_crs` resolves to its declared CRS, and whose CRS displacement is 0.0 m — it returns `2.167e-12`. The implementation converts registration coordinates into the candidate CRS's own geographic coordinate system (`CloneGeogCS`) and compares their extrema with the sheet's graticule metadata, or with quarter-degree snapping when those metadata are absent. That calculation does not independently test the subsequent transformation to WGS 84. It can reject inconsistent registration while remaining insensitive to the omitted datum transformation at issue here. The ∅ in Table 1 denotes this fault's inability to disturb the comparison; it does not mean the check must return zero when other errors are present.

The same check failed a second time, in the opposite direction, and the second failure is worth recording because it is the first one's mirror. `graticule_error` read its transformed coordinate pair positionally, and the two spatial-reference objects it compares disagree about axis order while reporting the same axis strategy. Every sheet the datum correction moved was therefore transposed, producing an apparent error of tens of degrees, and **370 of 437 sheets were rejected — the datum fix thrown out by the check it was meant to supersede**. Pinning both operands to traditional order took the rejections from 370 to 1. A check that cannot see a real fault and a check that rejects a real correction are the same defect seen from two sides: in neither case was the quantity being compared the quantity the reader assumed.

The fault is also not uniform across the series, and its shape is geographic rather than random. Figure 6 places all 627 indexed cells and colours each by the route its sheet took. The 269 displaced sheets fall in two groups, north and south, separated by a band between roughly 14°N and 17°N in which no sheet took the faulty route at all. That band is why a single probe can mislead: the archive holds two spatial populations, not one translated whole.

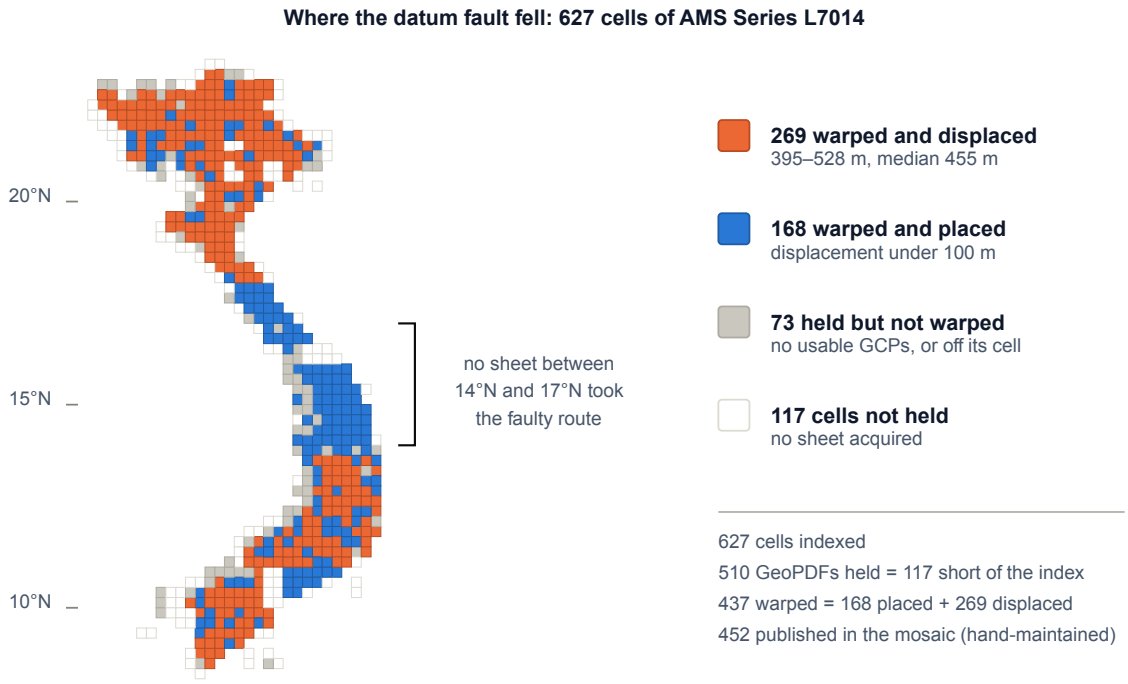


Figure 6: The fault has a geographic boundary. Each square is one indexed L7014 cell. Orange marks 269 displaced sheets; blue marks 168 placed sheets. No sheet between 14°N and 17°N took the faulty route. A seam can expose the fault where these populations meet; neighbouring sheets that share a displacement can still agree. Grey and empty cells distinguish missing warp results from missing holdings. Counts describe the controlled reproduction; 452 is the archive's hand-maintained publication claim.

7.5 CRS selection against an adopted frame

The corrected route adds `pick_crs` before the graticule check. It scores the declared or fallback CRS and, where available, an Indian 1960 alternative with explicit Everest 1830 (1937 Adjustment) Helmert parameters. For each candidate, it transforms the registration coordinates to WGS 84 and averages, over the expected cell corners, the distance to the nearest transformed registration point. Registration coordinates come from embedded GCPs, with neatline vertices as the fallback. It selects the candidate with the smaller score only if that score is at most 150 m. If every available candidate exceeds that threshold, it rejects the sheet. If no score can be computed, however, the implementation retains the declared or fallback CRS; this case is not verified by the lattice decision. The 150 m threshold was set as a generous ceiling against gross datum or projection mismatches, not derived from the ~15 m index-to-outline agreement scale reported below; the two figures answer different questions — 150 m gates a candidate CRS in or out, 15 m describes how tightly an accepted candidate then agrees — and we have not established what a tighter `pick_crs` threshold would cost in false rejections.

The reference is external to an individual GeoPDF declaration, but it is not independent of the adopted datum transformation. `cell_corners` transforms the index coordinates with the same explicit Helmert parameters used by the alternative candidate. Agreement therefore tests consistency with the adopted frame and can expose a missing transformation, while a shared error in those parameters could remain undetected. The result does not establish absolute accuracy at the approximately 15 m scale of the reported index-to-outline agreement. Testing that claim would require appropriately distributed reference positions whose coordinates were not generated by the same transformation.

This distinction preserves the useful result without giving the selection rule a broader authority than its inputs support. Refusal catches cases outside the scored alternatives; agreement identifies a compatible interpretation conditional on the adopted frame.

7.6 The fix, validated without being applied

The faulty and corrected passes compared below isolate exactly one pipeline flag (`l7014_mosaic.py warp --no-datum-shift`); no other stage of the build changed between them, so the before/after difference reported here is attributable to the datum-transformation step alone. This correction has been verified only by the same author and pipeline that produced the original datum-blindness fault; it has not yet been independently reproduced by a third party. Applying the explicit Helmert transformation in a dry run changes which sheets reach their cell at all. The corrected pass warps 436 sheets to the faulty pass's 437, and 434 of them land on their cell against 161 before. Sheets more than 150 m off their cell fall from 276 of 437 to 2 of 436. The median miss among on-cell sheets barely moves — 11 m before, 10 m after — because the correction does not improve sheets that were already placed; it moves displaced sheets onto their cells. For the same reason the worst on-cell miss rises, from 50 m to 95 m, with a net increase of 273 on-cell sheets across the two builds. The two the correction leaves off cell are `6630-4` Xa Phan Thiet (mean 662 m) and `6349-4` Cua Tra Ly (mean 396 m); in both the worst corner far exceeds the mean, which reads as a distorted or mis-clipped outline rather than a datum problem. The same method leaves the 24 hand-georeferenced sheets outside the fault: their control-point residuals remain 2.3–19.0 m RMS. These figures validate the proposed interpretation and correction; they do not describe a rebuilt public mosaic.

That distinction is deliberate. The `l7014-20260913` archive was never re-warped or re-

uploaded, and while it was served its state was worse than ”uncorrected”: its sheet manifest returned 404, because the build shipped pixels without the accompanying `.geojson`, so `fit` could not be run against the serving layer at all and the 452 constant quoted in §3 could not be checked against it. What readers received was therefore not merely displaced but unmeasurable from outside — itself an instance of §7.7's point that a serving layer needs its own instrument. The archive stopped being served on 2026-10-03, when the series moved to one IIIF image per sheet, each placed by its own georeference annotation; that layer is not measured here. Its bytes were removed from public storage on 2026-10-29 and are kept offline by the author. The published fault is therefore part of the result, not a historical problem retrospectively erased from the evidence. Reporting the dry run separately prevents it from being mistaken for a measurement of any serving layer.

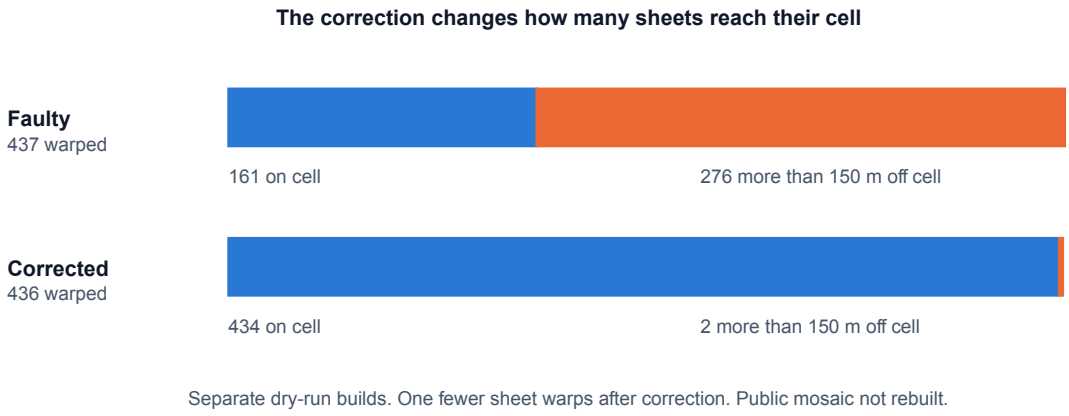


Figure 7: The correction recovers cell placement in the local reproduction. Bars use a common sheet-count scale. The faulty build places 161 of 437 sheets on cell; the corrected build places 434 of 436. Orange marks sheets more than 150 m off cell. The two remaining failures require review. These results do not measure a rebuilt public mosaic.

7.7 One layer down: a serving check

The same scope rule applies after georeferencing, when a correct source image is requested from an image service. An audit of the Indochine IIIF endpoint found that its generated tiles use explicit `w,h` size segments, whereas canonical IIIF clients, including the rendering stack used here, request a width-only segment. The two forms agree at full resolution, so a superficial whole-image test passes. One level below the overview they disagree for 58.6% of tiles across the 83-sheet survey. The resulting 404 reads as “Source not found”, which describes neither the source nor the real fault: the endpoint cannot compute an arbitrary requested size at that scale.

This is not a georeferencing error and should not inflate the positional claims above. It belongs in the taxonomy because it demonstrates the same operational lesson at the serving layer. A successful full-resolution request does not test the tile keys a viewer will actually request. The appropriate instrument is a tile-key audit against keys present in the object store; it has no opinion about datum, lattice or seam geometry, and those checks have no opinion about it.

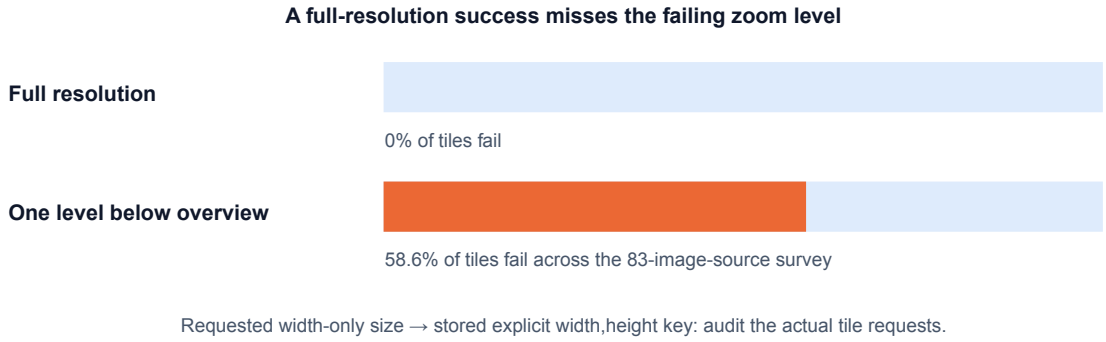


Figure 8: Test the resolution that the viewer requests. Orange is the share of tile requests that fail at the stated level. Full-resolution success coexists with 58.6% failures one level below the overview in the 83-image-source survey. This measures serving behaviour, not positional accuracy; no intermediate zoom-level rates are inferred.

7.8 The same structure in an automated pipeline

The 1:100,000 pipeline (§6) was built after the L7014 fault was understood, and it still has a shared input. Its per-sheet gates compare the detected frame with the catalogue box: the frame's pixel aspect against the box's ground aspect, the two ground-scale estimates against each other, and the four rim offsets against one another. A series pass adds the rim offset against the series median and footprint overlap between neighbouring slots. The rim-offset gates read only the image, so they bear on frame detection, not placement. The gates that bear on placement read the catalogue box, and the catalogue box is also what places the sheet. They can therefore see an index error only when it distorts the box's shape.

Tri Binh (W) is that case. Its detected frame is clean: four rim offsets of 169–172 px, fit residuals of 2.2–3.8 px, and 0.69% shape error against its catalogue box. Every per-sheet gate passes it. When its printed ticks were read again, three per axis, the catalogue's east and north edges agreed with the print to within reading noise (−59 m and +50 m). Its west and south edges were wrong by +509 m and +544 m. That is about 1.3% of the sheet's width and 1.1% of its height: the box is too large on both axes by similar proportions, so its shape barely changes. The error is comparable to the L7014 displacement (395–528 m), and the gates could not report it. Under the corrected box the shape error falls to 0.43%, which is consistent with the correction being real. The sheet was found only because it had been drawn into the calibration sample, the one step where the print, rather than the catalogue, supplies coordinates. There the calibration gate flagged it as an outlier, and its exclusion was recorded as a judgement, not a finding. The re-reading shows it was neither a regional datum break nor a misread tick, but an error in the index.

Quang Ngai marks the other side of the boundary. Its catalogue east edge was off by about 9.2 km, enough to push its declared longitude span to 0.5235 grades, outside the series band, while its detected frame's pixel aspect implied about 0.428. The span gate held it before any tick was read; the printed ticks later gave 0.4281 grades. A shape-changing index error is visible to the gates, and a shape-preserving one is not. This is the § mark in Table 1.

The frame detector also has a blind spot of its own. Two sheets were once landed with a boundary on a kilometre-grid line about 52 px inside the true neatline. The ruled line fit with 0.28–0.48 px residual across 24 patches; the engraved frame fits at about 3.2 px. A straighter fit is not a more correct one, so the residual could not flag the choice (¶ in Table 1). What caught it was a per-pixel profile of where the blank paper ends. That tests what the line bounds

rather than how straight it is, which is a different property from the one the fit measures.

These are cases, not rates. At the recorded state 314 of the 360 sheets are public and the series lattice check reports 328 clean placements with no conflicts. Printed ticks, however, were read only on the calibration sample and on sheets the gates had already held. How many public sheets carry an error like Tri Binh's is therefore unknown. The detector itself reached its 39th version within a week, each change justified on a handful of inspected sheets. That is a development record, not a held-out evaluation. §9 describes the measurement that would supply a rate.

7.9 Testing the table by fault injection

Table 1 is argued from each check's inputs and from one incident per cell. That is a claim about mechanism, not a measurement, so for the four checks this paper runs we tested the L7014 cells by injecting faults whose size is known by construction and running the checks unchanged (`scripts/l7014/paper_fault_injection.py`; output `docs/paper/fault-injection.json`). The predictions were written in the script before its first run and are graded against it. Thirty sheets were drawn with a fixed seed from the corrected build, after requiring a clean baseline of each (graticule error within tolerance, lattice error under 30 m); one drawn sheet failed that requirement and was replaced. Four faults were injected at 25, 50, 100, 150, 300, 455 and 800 m: a datum translation (the Helmert scaled so the displacement has the stated size, along the direction of the real fault); a misregistration of one sheet's control points with the frame correct; a sheet filed under a neighbouring cell; and, once, the WGS 84 ellipsoid substituted on the declared projection. The seam census was run on the whole series with 0, 10, 50 and 100% of sheets displaced, five random subsets for the partial fractions, and on a single displaced sheet. Detection uses each check's own threshold: $5\times10^{\circ}$ for `graticule_error`, 150 m for the lattice residual and `pick_crs`, and the 100 m seam cutoff used in §7.3.

fault	check	silent	fires	what was measured
datum translation	`graticule_error`	every size	never	identical to the baseline within 10 ° at all seven sizes
datum translation	lattice residual	100 m	300 m	28 of 30 sheets at 150 m
datum translation, all sheets	seam census	every size	never	largest edge median 72.4 m before, 73.5 m after an 800 m shift
datum translation, 10% or 50% of sheets	seam census	25 m	100 m	at 50 m, 1 of 5 and 2 of 5 subsets
ellipsoid substituted	`graticule_error`, lattice	never	30 of 30, both	
one sheet misregistered	`graticule_error`	50 m	100 m	
one sheet misregistered	lattice residual, `pick_crs`	100 m	300 m	10 of 30 at 150 m
one sheet misregistered	seam census	50 m	150 m	2 of 5 at 100 m
sheet filed under a neighbouring cell	`graticule_error`	every case	never	identical to the baseline
sheet filed under a neighbouring cell	lattice residual	never	30 of 30	

Every prediction held, and three results sharpen the table. First, the ∅ in the datum column is exact for a translation and does not extend to the datum as a whole: substituting the WGS 84 ellipsoid on the same projection moves the inverse-projected coordinates, and `graticule_error` rejected all 30 sheets. The existing `check` phase asserts the same thing, so the claim of §7.4 is a claim about the translation, and Table 1 now says so. Second, a common displacement is silent to the seam census at every size, as §7.3 states: this is measured, not only argued. The census fires only when the displaced sheets border undisplaced ones, and then from about 100 m. Third, the checks cover different ranges of one fault. A misregistration of 55–150 m is seen by `graticule_error` and by neither the lattice residual nor `pick_crs`, whose 150 m tolerance was chosen by inspection; above about 300 m all three agree. The 150 m figure is therefore a statement about which faults `pick_crs` can refuse, not a calibrated error bound.

The limits are the ones that decide how much this supports. The faults are synthetic and are injected into one series by the author who wrote the checks. "Truth" is the adopted Helmert and the lattice built from it, so this measures what each check can detect relative to that frame, not absolute accuracy. The thresholds are the code's own, so where a cell sits at a threshold (150 m for the lattice residual) the result describes the threshold. The 100 m seam cutoff was fixed from the same build whose baseline is used here, so a zero false-alarm rate at no injected fault holds by construction. Two faults in the harness itself, a sheet outline reduced to its first ring and a "neighbouring cell" that was the sheet's own, were found because the results disagreed with the predictions; they are fixed, and they are a reason to expect the same of any reader's harness. The rows for the published methods, the Indochine collision and the 1:100,000 index-box error (§7.8) are not tested here and remain argued from scope.

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8 Negative results

Several plausible checks did not yield evidence strong enough to carry the argument. For the Indochine datum, we warped one 1903 sheet from its detected corners and compared the Red River with modern imagery. Across 794 rows the modern channel lay at a median of 58 m east of the historical one, with a ±180 m spread. That is compatible with a small east–west displacement, but the channel runs north–south and therefore provides no north–south constraint; a two-dimensional correlation peaked at the edge of its search window. River migration, not the georeference, dominates the remaining variation. The southeast confluence agrees to about 100 m, which is reassurance rather than a measurement.

Toponym lookup was less useful still. Of twenty village names transcribed from the sheet, only four returned a single OpenStreetMap result, and even an apparently good administrative match can describe a modern settlement rather than its historical core. Correlating historical ink density with modern red-roof pixels produced a smooth, peakless surface driven by the rectangular overlap of the masks. Neither result is retained as a weak positive. Both delimit what an external check would need before it could adjudicate the Indochine datum.

The image-derived method also has known exceptions. Gia Bình, Phúc Nhạc, Quất Lâm and Thái Bình need hand-placed corners because thin-frame editions or inconsistent rim offsets defeat the thick-line assumption. The unresolved pair of rim lines is worth roughly 30 m, and only the printed graticule ticks can settle it. These limitations are not failures of the lattice or seam instruments; they are limits on the quality of the per-sheet readings those instruments are asked to compare.

9 Discussion

The contribution is not a new universal accuracy statistic. It is a way to decide what a statistic can mean before reporting it. Every verification check has a scope, and some checks reuse a datum, frame or adjacency relation that the georeferencing procedure already assumed. Such a check may be entirely appropriate for detecting local misregistration while being incapable of detecting a shared displacement. The useful question is therefore not whether a residual is small in isolation, but which error classes could have made it small. This is a structural instance of the test oracle problem in software testing: an oracle is only as independent as the process that supplies it, and a relation-based check — asking whether an output is *consistent* rather than *correct* — inherits that limitation whenever the relation is derived from the same artefact it verifies (Barr, Harman, McMinn, Shahbaz & Yoo, 2015).

For a regular series, this yields a practical ordering of checks. Begin with low-cost relations the series itself supplies: lattice spacing, one-to-one cell occupancy and free seams. Use reference positions independent of the adopted transformation when absolute placement is the claim to be tested. The `pick_crs` decision provides a narrower consistency check (§7.5). Do not turn a diagnostic relation into a hard constraint if it will later be the only available test of that relation. And preserve refusal as an outcome: an anomalous sheet is information for review, not an invitation to publish the least implausible transformation. This ordering is offered retrospectively: the build that shipped the fault did not follow it, and the fault is the reason the ordering exists rather than evidence that a project working from it in advance would have caught this class of error unaided.

The failure was also organisational, not only algorithmic. The graticule check and the CRS-selection step were written and reviewed as separate units of work, each locally reasonable, without anyone asking whether the second step's inputs already depended on an assumption the first step existed to test. Nothing in the code review process flagged the shared Helmert parameters as a shared premise, because no step in that process asked what a passing check had already assumed. The algorithmic mechanism in §7.4 explains why the check could not have caught this fault; it does not by itself explain why a check with that structure was the one that shipped. That second question is a process question, not a geometry one, and it is the reason the ordering above is stated as a checklist rather than left implicit in the method.

The third case says what an automated pipeline needs before its clear set can carry a rate. The gates' committed input is the catalogue, so the independent evidence must come from somewhere the catalogue does not reach. Two sources are available without new instruments. One is the sheet's own printed ticks, read on a random sample of *cleared* sheets rather than only on held ones. The other is a person's judgement of what the catalogue cannot supply: whether roads and rivers continue across seams with already-placed neighbours, and whether a named landmark falls where the sheet puts it. That review must record two separate verdicts, whether the boundary is on the printed edge and whether the sheet is in the right place. A frame-only review would pass Tri Binh, reproducing in the review the shared input the gates already have. The approval should also be bound to the corners it approved, so that a later detector version cannot publish new geometry under an old decision. We have not yet run this audit. Until it is run, the 1:100,000 clear set carries the gates' scope and no measured error rate.

The reasoning may apply beyond these series where its structural conditions hold. A seam census requires enough adjacent sheets, a lattice requires a known regular scheme, and neither substitutes for an independent absolute reference. Not every series in the archive meets those conditions. A 1971 province atlas prints a common military grid but gives each province

its own frame, extent and layout, so it has a grid and no sheet lattice. A set of colonial arrondissement plans has neither. The first can take grid intersections as dense per-sheet control, and neighbouring provinces overlap rather than meet at an edge, so there is no lattice and no seam. The second offers only content matching or hand placement. Both are outside what this paper's lattice and seam instruments can verify. Conversely, an individual sheet can receive only per-sheet checks, which is precisely why its residual must not be made to imply more. The archive model in §4 is part of the method here: series membership, editions, gaps and serving routes are not merely catalogue metadata when they determine which verification relations can be computed.

10 Conclusion

A georeferenced map series arrives with more evidence about itself than a single sheet can offer, and the practical question is not how much of it to gather but what each piece is still free to say. A controlled reproduction of the archive's faulty route retained 437 sheets, including 269 with CRS displacements of 395–528 m that passed the graticule check. That check cannot independently validate the transformation from the sheet's geographic CRS to WGS 84. On the same 715 PDF/PDF edges, correction reduced the number above 100 m from 189 to zero. The comparison establishes improved outline agreement. CRS selection against the lattice supplies a separate test of consistency with the adopted Helmert transformation, whose absolute accuracy remains to be independently assessed. These build measurements do not establish the exact population or accuracy of the serving archive. The third case shows the structure again in a pipeline built with the first two in view: a 509–544 m index error passed every per-sheet gate, because the gates and the placement read the same catalogue.

The generalisable part is not the magnitude, which is particular to this archive, nor the instruments, which are not new. It is the bookkeeping: a verification check has a scope and a set of inputs it has already committed to, and those two facts determine which error classes it can still report. Table 1 is that bookkeeping for the checks we use and for three published ones, and every ∅ in it is the ordinary consequence of a sound design — including our own, which is the worst row in the table. Stating a residual without stating its scope is therefore not a small omission. It is the omission that let a 455 m error render, pass its check, and ship.

11 Data and code availability

The archive's derived data — georeference annotations, sheet and printing indexes, verification outputs and code used to produce them — will be deposited under CC-BY-4.0 in a versioned Zenodo release before submission. The release DOI is intentionally not fabricated in this draft and will be inserted only after the archive record is minted. Source scans are not redistributed: they remain under their holding institutions' terms and are cited by their IIIF resources or catalogue records. This separation preserves a reproducible account of the derived geometry without asserting rights over the underlying scans.

The implementation includes the L7014 mosaic pipeline, the Indochine frame and corner-reading pipeline, the 1:100,000 frame, calibration and placement pipeline (`scripts/indochine100k/indochine`, the series-index migrations, and the checks described in §7. The figures ledger records the source and measurement date of every number quoted in this paper. The serving archive is also versioned separately from a proposed correction: reproducibility requires readers to distinguish the evidence measured on the current layer from the dry run that supports its

rebuild.

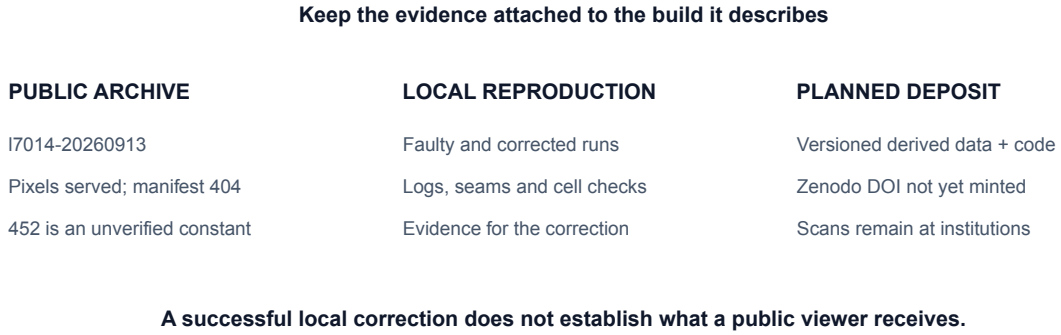


Figure 9: Three states of the evidence at the recorded audit. The served archive, local experiment and planned deposit have different evidentiary roles. Only the local reproduction supports the before/after counts. A future deposit must identify its build and preserve those distinctions; a DOI has not yet been minted.

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12 References

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